Version 1.0; Publication date: 2026-11-01
COSI — Corrugated Fault Codes

VDW
Die Wellpappenindustrie



COSI 1.0
Corrugated Standard Interface

Open industry standard for machine data exchange
in corrugated cardboard manufacturing

Imprint

COSI — Corrugated Fault Codes

COSI — Corrugated Standard Interface

Version	1.0 (prepublication)
Updated	2026-10-16
Publication date	2026-11-01
Status	Prepublication — package approved
Scope	The closed `faultCode` value set

Publisher / working group

VDW (Association of the German Corrugated Cardboard Industry e.V.)

Authors

Michael Sachs

Robert Schmid

Daniel Malolepszy

Dennis Strasser

Tobias Winkler

Technical editors

Technical specification development and quality assurance.

André Seifert (ePS)

Robin Boeckxstaens (OMP)

Contact

Christiane Trumpp

transparente-daten@vdw-da.de

Disclaimer

The PDF edition is a reproduction of the Markdown source published in the official COSI GitHub repository. In the event of any discrepancy, the Markdown source for the corresponding document version shall prevail.

https://github.com/vdw-da/COSI-Corrugated-Standard-Interface

Impressum / Legal Notice

Publisher: **Verband der Wellpappen-Industrie e.V. (VDW)**, a registered association. Registered address: Markgrafenstraße 19, 10969 Berlin, Germany. Represented by the Chair of the Board, Dr. Steffen P. Würth. Association register: VR 40431 B, Amtsgericht Charlottenburg (Charlottenburg Local Court). VAT identification number: DE 111 607 830. Telephone: +49 30 7543880-0. Email: info@wellpappen-industrie.de.

These publisher details are drawn from VDW's official Impressum — Legal Notice. For COSI questions and suggestions: transparente-daten@vdw-da.de. The repository's Apache License 2.0 governs COSI licensing; website-specific copyright restrictions are not imported into this repository.

Privacy

VDW's full privacy policy: Datenschutzerklärung — Privacy Policy.

Use of GitHub and sharing of content

We use the GitHub platform to provide, manage and collaborate on digital content. For this purpose, selected repositories, files, documentation and project-related information within our GitHub organisation may be made available to external users. Personal data (such as GitHub usernames and email addresses) may be processed where they are included in the shared content, arise through use of the platform or are needed to grant access. Processing serves to provide and share the relevant content and to collaborate with external users.

The platform provider is GitHub, Inc., USA. For details of how it handles your personal data, please see the GitHub Privacy Statement.

The company is certified under the EU–US Data Privacy Framework (DPF). The DPF is an arrangement between the European Union and the United States intended to ensure compliance with European data protection standards when data are processed in the United States. Each DPF-certified company undertakes to comply with these standards. Further information is available in the provider's DPF entry.

Inquiries by email, telephone or fax

If you contact us by email, telephone or fax, we store and process your inquiry, including all personal data it contains (name, inquiry), to handle your request. We do not disclose these data without your consent.

If your inquiry concerns documents or content provided through GitHub, it may be forwarded to the responsible working groups where this is necessary to assess or further process the inquiry. Inquiries that are not pursued further are deleted.

These data are processed on the basis of Article 6(1)(b) GDPR where your inquiry relates to the performance of a contract or is necessary to take steps prior to entering into a contract. In all other cases, processing is based on our legitimate interest in effectively handling inquiries addressed to us (Article 6(1)(f) GDPR), or on your consent (Article 6(1)(a) GDPR) where requested; consent may be withdrawn at any time. We retain data you send us through contact inquiries until you ask us to delete them, withdraw your consent to their storage, or the purpose for storing them no longer applies (for example, once your request has been dealt with). Mandatory statutory provisions, particularly statutory retention periods, remain unaffected.

3-letter failure mode codes for corrugated manufacturing, adapted from the failure-mode notations of **ISO 14224:2016, Annex B**. A non-null `faultCode` uses a code from this table; `internalCode` can carry optional vendor-specific detail. The maintenance measurement schema permits `faultCode: null` when the measurement is within limits.

Closed set for v1.0. The 40 codes below are the complete permitted non-null value set for `faultCode`, enforced as an `enum` in `schemas/maintenance-measurement.schema.json`. Adding a code requires a version change of this standard. Use `OTH` when a fault is not classifiable and `UNK` when the cause is unclear.

Code	Description	Examples
AIR	Abnormal instrument/sensor reading	Incorrect moisture value, faulty encoder, photoelectric switch dirty
BRD	Breakdown / major failure	Main drive failed, bearing seized, gearbox damage
CLW	Inter-line communication / signal fault	PLC communication disturbed, bus error, no signal from upstream/downstream
DOP	Delayed operation	Sluggish heating control, delayed vacuum/drive start
ELP	External leakage – process medium	Glue leak, steam/condensate leak, oil on web
ELU	External leakage – utility medium	Pneumatic leak, hydraulic leak, cooling water loss
ERO	Erratic/unstable output	Web oscillates, speed hunting, register drift
FOF	Faulty output frequency / speed	Motor control unstable, line speed fluctuates
FOV	Faulty output voltage / supply	Undervoltage drives, spikes, faulty PSU
FTC	Failure to close on demand	Guard/slide/valve won't close, glue valve won't shut
FTF	Failure to function on demand	Tool change won't start, no vacuum, print unit won't engage
FTI	Failure to function as intended	Poor cutting, creasing too shallow/deep, uneven glue
FTO	Failure to open on demand	Flap won't open, print unit won't disengage, gap adjust stuck
FTR	Failure to regulate/control	Web tension control not working, temperature drift, register control ineffective
FTS	Failure to start on demand	Aggregates don't start, line stuck in start mode
HIO	High output	Overspeed, excessive web tension, pneumatic overpressure
ILP	Internal leakage – process medium	Glue backflow, steam valve internally leaking
ILU	Internal leakage – utility medium	Valve internally leaking, hydraulic block bypass
INL	Internal leakage (general)	Leak inside aggregate without clear medium
LCP	Leakage in closed position	Glue valve drips when closed, steam valve passes
LOO	Low output	Vacuum too weak, pressure too low, drive power missing
NOI	Noise	Rubbing in print unit, whistling pneumatics, rattling chains
OHE	Overheating	Hot bearing, heater zone above setpoint, motor/drive overtemp
OTH	Other	Not classifiable / special case
PCL	Premature closure	Safety shutdown closes hood, glue valve closes sporadically
PDE	Parameter deviation	Moisture high/low, temperature out of tolerance, web tension too high
PLU	Plugged/choked	Glue filter blocked, ink chamber clogged, extraction blocked
POW	Insufficient power/energy supply	Air pressure low, steam supply drops, mains power missing
PTF	Power/signal transmission failure	Sensor cable break, slip ring signal lost, valve not actuated
SER	Minor in-service problems	Contamination, loose bolts, worn scrapers
SHH	Spurious high alarm level	False web break alarm, moisture high alarm without cause
SLL	Spurious low alarm level	Low pressure alarm despite OK, vacuum alarm sporadic
SLP	Slippage	Belt slip, web slips on roll, conveyor chatter
SPO	Spurious operation	Unwanted print unit opening, die cutter triggers without order
STD	Structural/mechanical deficiency	Damaged creaser wheel, broken knife, worn roll, corrosion
STP	Failure to stop on demand	Line runs on, aggregate ignores stop
UNK	Unknown	Cause unclear / incomplete message
UST	Spurious stop	Emergency stop without cause, safety chain trips sporadically
VIB	Vibration	Roll imbalance, print unit vibration, die cutter knocking
VLO	Very low output	Almost no glue, vacuum near zero, barely visible print

Imprint and Privacy Notice (Impressum / Datenschutz)